Prof. Ir. Dr. Wan Azlina Wan Ab Karim Ghani

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Head, Sustainable Process Engineering Research Centre (SPERC)

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Summary

Prof. Ir. Dr. Wan Azlina Wan Ab Karim Ghani is a chemical engineer at Universiti Putra Malaysia who turns waste and biomass into energy, carbon materials and practical environmental solutions. Her group built a 10 kW integrated mobile gasifier and the Biochar Microwave Carbonizer, and modelled Malaysia's oil-palm value chains in the Newton–RCUK BEFEW nexus project with UKM, UTP, UNMC and the University of Bath. She serves on the Department of Environment's EIA review panel and Standards Malaysia's TC 238 Solid Biofuels committee, and brings circular-economy thinking to schools and communities through STEM and SULAM service-learning.

Education

2005	PhD Combustion Engineering, University of Sheffield, UK
2000	B.Eng (Hons) Chemical Engineering, Universiti Teknologi Malaysia

Appointments and leadership

2025–2028	Head, SPERC, Faculty of Engineering, UPM
c. 2021	Research sabbatical, AAIBE Chair for Renewable Energy, UNITEN
Oct 2021–Feb 2024	Deputy Dean (Academic), Faculty of Engineering, UPM
2020–present	Professor, Department of Chemical and Environmental Engineering, UPM
2019–2021	Head, SPERC, Faculty of Engineering, UPM
2015–2017	Research Coordinator, Department of Chemical and Environmental Engineering, UPM
2012–2020	Associate Professor, Department of Chemical and Environmental Engineering, UPM
2012–present	Academic Coordinator, Department of Chemical and Environmental Engineering, UPM
Jun 2008–Mar 2012	Postdoctoral/Visiting Scholar, University of Melbourne
2007–2012	Senior Lecturer, Department of Chemical and Environmental Engineering, UPM
May 2005–May 2007	Lecturer, Department of Chemical and Environmental Engineering, UPM
Sep 2000–Apr 2005	Tutor, Department of Chemical and Environmental Engineering, UPM

Scopus snapshot

Documents	Citations	h-index	Q1 papers
150	7,103	38	58

Scopus author 57203833103, as of 11 October 2026. Google Scholar figures are not mixed in.

Research grants

RM 2,107,796 as principal investigator across 25 grants; RM 5,551,278 in total across 45 grants including co-researcher roles.

Largest grants as principal investigator

2017	Understanding biomass value chains and the environment-food-energy-water nexus in Malaysia through whole-system analysis and optimization. Private/international grant. RM 505,600
2014	Fabrication of a 10 kW integrated mobile gasifier for sustainable power and biofuels production. Prototype Research Grant Scheme (PRGS). RM 165,000
2009	Performance studies of anaerobic digestion of biowaste for better municipal solid waste management in Malaysia. Research University Grant Scheme (RUGS). RM 160,000
2016	Activated carbon monolith-supported metal oxide catalyst for simultaneous SOx and NOx removal. Geran Putra Inisiatif Putra Berkumpulan (GP-IPB). RM 148,200
2014	Process integration and supply chain optimisation of a sustainable biorefinery. Long-Term Research Grant Scheme (LRGS). RM 125,600
2006	Gasification of agrowaste to produce biofuel. Strategic Research Fund (SRF), MOSTI. RM 105,796

Twenty most-cited papers

1. Mohd Salleh, M.A.M.; Mahmoud, D.K.; Ghani, W.A.W.A.; Azni, A. (2011). Cationic and anionic dye adsorption by agricultural solid wastes: A comprehensive review. *Desalination*, 280, 1–13. doi:10.1016/j.desal.2011.07.019 1,802 citations
2. Ghani, W.A.W.A.K.; Mohd, A.; da Silva, G.; Rashid, U.; Al-Muhtaseb, A.H. (2013). Biochar production from waste rubber-wood-sawdust and its potential use in C sequestration: Chemical and physical characterization. *Industrial Crops and Products*, 44, 18–24. doi:10.1016/j.indcrop.2012.10.017 393 citations
3. Mahmoud, D.K.; Mohd Salleh, M.A.M.; Ghani, W.A.W.A.; Azni, A.; Abidin, Z.Z. (2012). Batch adsorption of basic dye using acid treated kenaf fibre char: Equilibrium, kinetic and thermodynamic studies. *Chemical Engineering Journal*, 181-182, 449–457. doi:10.1016/j.cej.2011.11.116 388 citations
4. Ghani, W.A.W.A.; Rusli, I.F.; Radiah, D.R.A.; Azni, A. (2013). An application of the theory of planned behaviour to study the influencing factors of participation in source separation of food waste. *Waste Management*, 33, 1276–1281. doi:10.1016/j.wasman.2012.09.019 336 citations
5. Raheem, A.; Ghani, W.A.K.G.; Yap, Y.H.; Danquah, M.K.; Harun, R. (2015). Thermochemical conversion of microalgal biomass for biofuel production. *Renewable and Sustainable Energy Reviews*, 49, 990–999. doi:10.1016/j.rser.2015.04.186 332 citations
6. Mohammed, M.A.A.; Salmiaton, A.; Ghani, W.A.K.G.; Mohd Salleh, M.S.; Ahmadun, A. (2011). Air gasification of empty fruit bunch for hydrogen-rich gas production in a fluidized-bed reactor. *Energy Conversion and Management*, 52, 1555–1561. doi:10.1016/j.enconman.2010.10.023 259 citations
7. Mohammed, M.A.A.; Salmiaton, A.; Ghani, W.A.K.G.; Ahmadun, A.; Yap, Y.H. (2011). Hydrogen rich gas from oil palm biomass as a potential source of renewable energy in Malaysia. *Renewable and Sustainable Energy Reviews*, 15, 1258–1270. doi:10.1016/j.rser.2010.10.003 238 citations
8. Chan, Y.H.; Cheah, K.W.; How, B.S.; Hong, B.H.; Ngan, S.L. (2019). An overview of biomass thermochemical conversion technologies in Malaysia. *Science of the Total Environment*, 680, 105–123. doi:10.1016/j.scitotenv.2019.04.211 185 citations

9. Haq, I.U.; Qaisar, K.; Nawaz, A.; Ghani, W.A.W.A.K.; Choong, T.S.Y. (2021). Advances in valorization of lignocellulosic biomass towards energy generation. *Catalysts*, 11, 309. doi:10.3390/catal11030309 162 citations
10. Mohammed, M.A.A.; Salmiaton, A.; Ghani, W.A.K.G.; Mohd Salleh, M.S. (2012). Gasification of oil palm empty fruit bunches: A characterization and kinetic study. *Bioresource Technology*, 110, 628–636. doi:10.1016/j.biortech.2012.01.056 153 citations
11. Rasit, N.; Azni, A.; Harun, R.; Ghani, W.A. (2015). Effects of lipid inhibition on biogas production of anaerobic digestion from oily effluents and sludges: An overview. *Renewable and Sustainable Energy Reviews*, 45, 351–358. doi:10.1016/j.rser.2015.01.066 140 citations
12. Harun, R.; Yip, J.W.S.; Thiruvenkadam, S.; Cherrington, T.; Danquah, M.K. (2014). Algal biomass conversion to bioethanol-a step-by-step assessment. *Biotechnology Journal*, 9, 73–86. doi:10.1002/biot.201200353 129 citations
13. Ngabura, M.; Hussain, S.A.; Ghani, W.A.W.A.; Jami, M.S.; Tan, Y.P. (2018). Utilization of renewable durian peels for biosorption of zinc from wastewater. *Journal of Environmental Chemical Engineering*, 6, 2528–2539. doi:10.1016/j.jece.2018.03.052 109 citations
14. Ahmed, A.M.A.; Salmiaton, A.; Choong, T.S.Y.; Ghani, W.A.K.G. (2015). Review of kinetic and equilibrium concepts for biomass tar modeling by using Aspen Plus. *Renewable and Sustainable Energy Reviews*, 52, 1623–1644. doi:10.1016/j.rser.2015.07.125 104 citations
15. Shaharum, N.S.N.; Shafri, H.Z.M.; Ghani, W.A.W.A.K.; Al-Habshi, M.M.A.; Yusuf, B. (2020). Oil palm mapping over Peninsular Malaysia using Google Earth Engine and machine learning algorithms. *Remote Sensing Applications: Society and Environment*, 17. doi:10.1016/j.rsase.2020.100287 87 citations
16. Ghani, W.A.W.A.K.; Alipour Moghadam Esfahani, R.A.; Mohd Salleh, M.A.M.; Alias, A.B. (2009). Air gasification of agricultural waste in a fluidized bed gasifier: Hydrogen production performance. *Energies*, 2, 258–268. doi:10.3390/en20200258 86 citations
17. Silas, K.; Ghani, W.A.W.A.K.; Choong, T.S.Y.; Rashid, U. (2019). Carbonaceous materials modified catalysts for simultaneous SO2/NOx removal from flue gas: A review. *Catalysis Reviews - Science and Engineering*, 61, 134–161. doi:10.1080/01614940.2018.1482641 79 citations
18. Abdullah, T.G.; Ghani, A.G.K.; Yunus, Y.; Omar, R. (2006). Biomass as the renewable energy sources in Malaysia: An overview. *International Journal of Green Energy*, 3, 323–346. doi:10.1080/01971520600704779 79 citations
19. How, B.S.; Ngan, S.L.; Hong, B.H.; Shuhaili, A.F.A.; Rambli, J. (2019). An outlook of Malaysian biomass industry commercialisation: Perspectives and challenges. *Renewable and Sustainable Energy Reviews*, 113. doi:10.1016/j.rser.2019.109277 76 citations
20. Motlagh, S.R.; Harun, R.; Radiah, D.R.A.; Wilfred, C.D.; Elgharbawy, A.A.M. (2019). Screening of suitable ionic liquids as green solvents for extraction of eicosapentaenoic acid (EPA) from microalgae biomass using COSMO-RS model. *Molecules*, 24, 713. doi:10.3390/molecules24040713 65 citations

Postgraduate supervision

	PhD, Main supervisor	PhD, Co-supervisor	MSc, Main supervisor	MSc, Co-supervisor
Graduated	8	14	11	8
Current	0	5	3	1

50 postgraduate students in total, 41 of them graduated (named list, Research group page).

Awards

2025	World's Top 2% Scientists (Stanford University–Elsevier), 2025 – Single-year impact
2024	World's Top 2% Scientists (Stanford University–Elsevier), 2024 – Single-year impact
2024	Invention, Innovation & Design Exposition 2024 (IIDEX 2024), Universiti Teknologi MARA (UiTM)
2024	3rd International Invention, Innovation, Technology Competition & Exhibition 2024 (ITEX 2024) – Advanced Graphene-Based Oil and Heavy Metal Absorbent for Industrial Wastewater Filtration, Malaysian Invention & Design Society
2023	World's Top 2% Scientists (Stanford University–Elsevier), 2023 – Single-year impact
2022	World's Top 2% Scientists (Stanford University–Elsevier), 2022 – Single-year impact
2021	Top Research Scientists Malaysia (TRSM) 2021, Academy of Sciences Malaysia
2021	World's Top 2% Scientists (Stanford University–Elsevier), 2021 – Single-year impact
2019	Biochar Microwave Carbonizer (BMC), ITEX 2019
2019	Green Biofilter System Award, Malaysian STEM Teachers Association
2019	Cluster Award Winner (Energy and Environment), Malaysian STEM Teachers Association
2019	Gold award, PENCIPTA 2019 – Biochar Microwave Carbonizer
2018	Venus International Women Awards (VIWA) 2018 – Outstanding Women in Engineering, Venus International Foundation (India)
2018	KANS Honours Diploma – Energy, Knowledge Application and Notion for Society (KANS), Mustafa Science & Technology Foundation (MSTF), 2018
2018	Silver medal, Minggu Penyelidikan dan Inovasi (MPI) 2018 – Eco enzyme for wastewater treatment, Universiti Malaysia Terengganu
2018	Silver medal, ITEX 2018 – Hydrogel Charsh (HC) as a powerful adsorbent
2018	Silver medal, I-IDEA 2018 – Image classification for mapping oil palm distribution via support vector machine
2017	Best Overall Knowledge Transfer Programme Project, MARS
2017	Asia-Pacific University-Community Engagement Network (APUCEN) / International Youth Volunteer Award (Environment Category), APUCEN Summit
2015	Presidential Award of Excellence, Institution of Engineers Malaysia (IEM)
2012	L'Oréal–UNESCO For Women in Science (FWIS) National Fellowship 2012, L'Oréal Malaysia / UNESCO

Intellectual property

2026	Ecobrick Furniture. Copyright, LY2025W02604, Registered.
2026	Dual-crosslinked PVA/bamboo-based cellulose hydrogel electrolyte. Patent, PI2026005760, Filed.
2025	A 3-Step Odor Neutralizer Recipe. Copyright, LY2025W12617, Notified.
2025	Upcycle Waste Paper Module. Copyright, LY2025W13666, Notified.
2025	SULAM: Sustainable Pet Clothes. Copyright, LY2025W13667, Notified.
2025	SULAM: Eco-Lip Balm. Copyright, LY2025W13668, Notified.
2024	Transforming Fish Waste to Pet Food. Copyright, LY202400129, Filed.
2024	DeMI Interface Tool for Profit Estimation and Waste Conversion Technology Recommendations. Copyright, LY202400913, Filed.
2019	Eco Enzyme for Wastewater Treatment. Copyright, LY2019000460, Registered.

2019	Production of Oil Palm Spatial Distribution Maps Via Computing Approach. Copyright, LY2019009020, Registered.
2016	Hydrogel charsh (HC) as powerful adsorbent. Copyright, LY2016002144, Registered.

Selected talks

2026	Keynote, 23rd Carbon Research International Forum (host: Universiti Teknologi PETRONAS)
2019	Invited plenary speaker, International Symposium on Technologies and Materials for Renewable Energy, Environment and Sustainability (TMREES 2019), Faisalabad, Pakistan
2019	Trainer, Training on sludge and wastewater treatment
2018	Invited speaker, ICBioE 2018
2018	Invited speaker, POMREQ 2018 (National Seminar on Palm Oil Milling, Refining, Environment and Quality)
2017	Invited speaker, Environmental pollution management
2012	Invited speaker, Seven keys to becoming a successful international researcher
2011	Invited speaker, Training on industrial wastewater treatment
2008	Invited speaker, Landfill management and design

Professional service

Editorial

2023	Associate editor, Waste and Resource Management (Emerald)
2025	External editor, Journal of Oil Palm Research (MPOB)

Expert panels

2017	Expert panel member, National Policies on Science, Technology and Innovation (NPSTI) study for ASM, Academy of Sciences Malaysia (ASM)
2019–present	Panel assessor, EIA report review, Department of Environment (DOE)
2019–2020	Technical advisory panel for project evaluation, Ministry of Housing and Local Government (MHLG)
2019	Invited panellist for Biasiswa Agung, Public Service Department (JPA)
2019	Invited panellist for Skim Latihan Persekutuan, Public Service Department (JPA)
2024–present	MQA accreditation assessor, MQA
	External assessor for academic promotion, Local and overseas universities
	Regional director, CRES

Compiled from the records on this website, 11 October 2026. Scopus, ORCID, Google Scholar